



Each of Edison's wax cylinders could hold around two minutes of audio.

The first records

- **What?** Sound recording
- **Who?** Thomas Edison
- **Where and when?** US, 1877

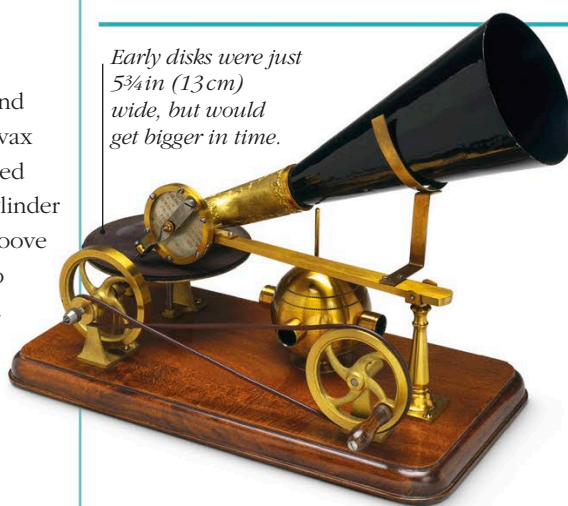
Edison discovered how to record sound in 1877. By 1888, he had developed wax cylinders that were the first widely used method of listening to music. Each cylinder was molded from hard wax with a groove on the outer surface corresponding to a sound recording. It could be played using a phonograph (see p.204). Early phonographs or “talking machines”, as they were originally called, were operated by clockwork.

FAST FACTS

- The first ever sound recording was of Thomas Edison reciting the nursery rhyme “Mary Had a Little Lamb.”
- Early disk records were made of shellac—a type of resin produced by insects.
- Record players didn’t become truly loud until the 1920s, after the invention of electric speakers by the American engineers Peter L. Jensen and Edwin Pridham in 1915.

Listen to the music

The devices we use for listening to music have changed dramatically over the years. For much of the 20th century, most of the developments were aimed at improving sound quality. In recent times, the move has been toward making equipment smaller and lighter. A record collection—along with the device to play the records—has evolved from something taking up shelves of space to something that fits in your pocket.



Early disks were just 5¾in (13cm) wide, but would get bigger in time.

Flat disks

- **What?** Berliner gramophone
- **Who?** Emile Berliner
- **Where and when?** US, 1887

Phonographs were popular but expensive, and took up a lot of space. The German-American inventor Berliner came up with a cheaper, less bulky alternative with his invention of a machine that traced sound grooves onto flat disks. These records were not only more durable, they were easier to produce (as they could be simply stamped out using a template) and store, and in their final form they also held more music—up to five minutes.

Long-playing (LP) records

- **What?** Vinyl 45 and 33 rpm records
- **Who?** RCA Victor (45s) & Columbia Records (33s)
- **Where and when?** US, 1948

Early disks revolved at 78 rpm (rotations per minute) when played. In 1948, two new longer-playing formats were introduced. Made of a plastic called vinyl, they played at different speeds. The 45 rpm extended play (EP) records could hold around four minutes of sound per side while the 33 rpm long-playing (LP) records could hold 25 minutes per side.



Vinyl record, 1971